

A COUNTEREXAMPLE TO KOUROVKA NOTEBOOK PROBLEM 16.45

ALEC KRIEBEL

Abstract

Let $b(G)$ be the largest size of an inclusion-minimal base over all permutation representations of a finite group G , with bases taken for the permutation image, and let $\mu'(G)$ be the largest size of an independent subset of G . We construct an explicit affine group of order 100920 for which $b(G) = 3 < 4 = \mu'(G)$, giving a negative answer to Kourovka Notebook Problem 16.45. The maximum remains three when restricted to faithful representations. The proof is structural: every subgroup missing the translation subgroup has faithful base invariant at most two, and every nontrivial normal subgroup contains the translations. The independent four-set gives an irredundant intersection family whose intersection is central in the complement but not normal in the affine group.

2020 *Mathematics subject classification*: Primary 20B05; Secondary 20D30, 20D60.

Keywords and phrases: Permutation bases, independent sets, finite groups, subgroup lattices, Kourovka Notebook.

1. Introduction

A subset S of a finite group X is *independent* if $x \notin \langle S \setminus \{x\} \rangle$ for every $x \in S$. Write $\mu'(X)$ for the largest size of such a subset; it need not generate X . For an action $\rho : X \rightarrow \text{Sym}(\Omega)$, a *base* is a set of points with pointwise stabilizer $\ker \rho$. A base is *minimal* if it is inclusion-minimal. Let $b(X)$ be the largest size of a minimal base over all permutation representations of X , and let $b_f(X)$ denote the same maximum restricted to faithful representations. In particular, $b(1) = b_f(1) = \mu'(1) = 0$. The definitions allow intransitive and nonfaithful actions. Allowing infinite Ω makes no difference: choosing a moved point for each nonidentity element of the finite image gives a finite subbase of every base.

Cameron's Problem 16.45 in the *Kourovka Notebook* asks whether $b(X) = \mu'(X)$ for every finite group X [4]. The inequality $b(X) \leq \mu'(X)$ is known. Cameron uses b_2 for b , and uses μ' in [2] and $\mu^\uparrow$ in [3] for our independence invariant. In contrast, Anagnostopoulou-Merkouri and Burness [1, Remark 1(c)] restrict their b_2 to faithful actions, so it corresponds to our b_f . The all-actions question is still posed in Cameron's 2024 account [3, Section 6, Proposition 2 and the following discussion] and is listed without a solution comment in the September 2026 notebook. We prove the following.

THEOREM 1. *Over $\mathbf{F}_{29}$, put*

$$A = \begin{pmatrix} 0 & 28 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 7 \\ 12 & 28 \end{pmatrix}, \quad H = \langle A, B \rangle \leq \mathrm{SL}_2(29).$$

For the natural affine semidirect product $G = \mathbf{F}_{29}^2 \rtimes H$,

$$|G| = 100920, \quad b_f(G) = b(G) = 3 < 4 = \mu'(G).$$

The proof separates two intersection conditions in the subgroup lattice. Independence corresponds to irredundant intersection families with arbitrary intersection, whereas permutation bases require that intersection to be normal. In our example an independent four-set gives a family with intersection of order two, central in H but not normal in G . The faithful upper bound follows by restricting an intersection family to a subgroup that does not contain the translations. The normal-subgroup structure then handles every nonfaithful action. No enumeration of the subgroup lattice of G is required, and no minimum-order assertion is made.

2. Subgroup formulations

We recall Cameron's subgroup formulations [2, Propositions 3.1–3.2 and Corollary 3.3]; see also [3]. We include the short arguments to make the normality condition explicit.

PROPOSITION 2. *The invariant $b(X)$ is the largest t for which subgroups $L_1, \dots, L_t$ of X satisfy*

$$N = \bigcap_{i=1}^t L_i \triangleleft X, \quad \bigcap_{j \neq i} L_j > N \quad (1 \leq i \leq t).$$

The same description with $N = 1$ gives $b_f(X)$. Consequently,

$$b(X) = \max_{N \triangleleft X} b_f(X/N). \quad (1)$$

PROOF. Point stabilizers of a minimal base have the stated properties. Conversely, let X act on the disjoint union of the left coset spaces X/L_i . The action kernel is

$$\bigcap_i \mathrm{Core}_X(L_i) = \mathrm{Core}_X \left(\bigcap_i L_i \right) = N.$$

The identity cosets form a minimal base. Factoring through the action kernel gives (1). Empty intersections are taken in the ambient group, so the empty family also handles the trivial group and trivial actions. $\square$

Call a subgroup family *meet-irredundant* if every member is essential to its total intersection, without a normality condition.

LEMMA 3. *The largest size of a meet-irredundant family in X is $\mu'(X)$. In particular, $b(X) \leq \mu'(X)$.*

PROOF. For a meet-irredundant family $L_1, \dots, L_t$, choose $x_i \in (\bigcap_{j \neq i} L_j) \setminus L_i$. Every x_j with $j \neq i$ lies in L_i , whereas x_i does not, so the x_i are distinct and independent. Conversely, for an independent set $\{x_1, \dots, x_t\}$, the subgroups $L_i = \langle x_j : j \neq i \rangle$ are meet-irredundant, with witnesses x_i . $\square$

LEMMA 4 (Restriction). *If $L_1, \dots, L_t$ is meet-irredundant in X , then $t - 1 \leq \mu'(L_i)$ for each i . If its total intersection is 1, then $t - 1 \leq b_f(L_i)$.*

PROOF. Inside L_i , the family $L_i \cap L_j$ for $j \neq i$ is meet-irredundant: the witness for L_j belongs to L_i . Its total intersection is unchanged. Apply Lemma 3 or Proposition 2. $\square$

These descriptions also show that μ' and b_f are monotone under subgroups. For completeness, a meet-irredundant family gives the top-preserving Boolean meet-semilattice embedding

$$J \mapsto \bigcap_{i \notin J} L_i \quad (J \subseteq \{1, \dots, t\}).$$

The witnesses distinguish the images, and the bottom is $\bigcap_i L_i$. Thus the normality distinction is exactly the one in the notebook's semilattice formulation; no preservation of joins is assumed.

3. The complement

Let H be as in Theorem 1, and set $Z = \{I, -I\}$. Direct multiplication gives

$$A^2 = B^3 = (AB)^5 = -I. \quad (2)$$

The spherical triangle presentation $A_5 = \langle a, b \mid a^2 = b^3 = (ab)^5 = 1 \rangle$ and simplicity of A_5 show that $H/Z \cong A_5$: the quotient is nontrivial because $A \notin Z$. Hence $|H| = 120$.

We recall the elementary subgroup facts that will be used. Every proper subgroup of A_5 lies in a subgroup isomorphic to A_4 , D_{10} , or S_3 , where D_{10} has order ten. Indeed, in the natural action on five letters an intransitive subgroup fixes a letter or has orbits of sizes $2 + 3$. A proper transitive subgroup has order 5, 10, 15, 20, or 30. Orders 15 and 20 would force a normal Sylow 5-subgroup, whose normalizer in A_5 has order ten; order 30 is excluded by simplicity. The remaining cases lie in a Sylow 5-normalizer.

Each of A_4, D_{10}, S_3 has independence invariant two. For A_4 , any generating set contains a 3-cycle and an element outside its cyclic subgroup; these generate A_4 , since it has no subgroup of order six. The proper subgroups of A_4 have independence invariant at most two. For a dihedral group of order $2q$ with q prime, a generating set contains either a nontrivial rotation and a reflection, or two distinct reflections; two such elements suffice. Deleting one element from an independent set in A_5 therefore gives

$$\mu'(A_5) \leq 3. \quad (3)$$

The natural degree-five action has a minimal base of size three, so equality holds. These facts also follow directly from the usual subgroup structure of A_5 ; simplicity follows from its nonidentity conjugacy-class sizes 15, 20, 12, 12.

LEMMA 5. *One has $\mu'(H) = b(H) = 3$ and $b_f(H) = 2$.*

PROOF. The only involution in $\mathrm{SL}_2(29)$ is $-I$: a matrix squaring to I is diagonalizable with eigenvalues ± 1 , and determinant one forces the signs to agree. Every even-order subgroup of H therefore contains Z . Every odd-order subgroup injects into A_5 and is trivial or cyclic of order three or five by the preceding subgroup description.

Let S be independent in H . If its images in H/Z , as an indexed family, are independent, then $|S| \leq 3$. Otherwise some $s \in S$ lies in $KZ \setminus K$, where $K = \langle S \setminus \{s\} \rangle$. Thus $Z \not\subseteq K$, so K is trivial or cyclic of odd prime order. Independence then gives $|S| \leq 2$. Hence $\mu'(H) \leq 3$. The degree-five action of H/Z pulls back to an action of H with a minimal base of size three. Lemma 3 proves $b(H) = \mu'(H) = 3$.

A subgroup family with total intersection 1 must have an odd-order member. If that member is trivial, a minimal such family has size one. If it has prime order, some other member intersects it trivially, so a minimal family has size at most two. Subgroups of orders three and five give a faithful family of size two. This proves $b_f(H) = 2$. $\square$

We will need an explicit independent triple in H . Put

$$R_1 = A, \quad R_2 = \begin{pmatrix} 12 & 24 \\ 0 & 17 \end{pmatrix}, \quad R_3 = \begin{pmatrix} 25 & 3 \\ 4 & 4 \end{pmatrix}. \quad (4)$$

The identities $R_2 = BAB^2$ and $R_3 = AR_2(AB)^2$ put these matrices in H . Direct multiplication gives $R_i^2 = -I$ and

$$\mathrm{ord}(R_2R_3) = 4, \quad \mathrm{ord}(R_1R_3) = 3, \quad \mathrm{ord}(R_1R_2) = 10.$$

In H/Z the corresponding products have orders 2, 3, 5. Each pair of projective involutions generates a dihedral group; lifting back to H , the pair subgroups have orders

$$|\langle R_2, R_3 \rangle| = 8, \quad |\langle R_1, R_3 \rangle| = 12, \quad |\langle R_1, R_2 \rangle| = 20. \quad (5)$$

The subgroup generated by all three has order divisible by $\mathrm{lcm}(8, 12, 20) = 120$, so it is H . Every pair subgroup is proper, proving independence. Moreover, the first two pair subgroups intersect in $\langle R_3 \rangle$: this subgroup has order four, which is also $\mathrm{gcd}(8, 12)$. The third does not contain R_3 and contains Z . Thus the intersection of the three pair subgroups is exactly Z .

4. Subgroups missing the translations

LEMMA 6. *Let p be an odd prime and let $C \leq \mathbf{F}_p^\times$ be a cyclic 2-group. For the faithful scalar semidirect product $L = \mathbf{F}_p \rtimes C$, one has $\mu'(L) \leq 2$.*

PROOF. Put $P = (\mathbf{F}_p, +)$ and consider a subgroup $J \leq L$. If $J \cap P = 1$, then J embeds in C and has independence invariant at most one: a subgroup of a cyclic prime-power group is generated by an element of maximum order in any generating set. Otherwise $P \leq J$ and $J = P \rtimes D$ for some $D \leq C$. For $D = 1$ the assertion is immediate. If

$D \neq 1$, every generating set of J contains an element $x = (a, \lambda)$ whose image generates D , because D has a unique maximal subgroup. Writing $m = |D|$, we have

$$x^m = \left(a \sum_{j=0}^{m-1} \lambda^j, 1 \right) = 1.$$

Thus $\langle x \rangle$ is a complement of order m . Some other generator y lies outside it. The subgroup $\langle x, y \rangle$ projects onto D ; if it intersected P trivially, its order would be m and it would equal $\langle x \rangle$, a contradiction. Hence $\langle x, y \rangle = J$. Every independent generating set of every subgroup of L has size at most two. $\square$

LEMMA 7. *Every line stabilizer in H is cyclic of order dividing four. In particular, H acts irreducibly on $V = \mathbf{F}_{29}^2$.*

PROOF. For a line W , the scalar action gives a homomorphism $H_W \rightarrow \mathbf{F}_{29}^\times$. In a basis adapted to W , a kernel element has the form $\begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix}$. If $u \neq 0$ it has order 29, impossible in a group of order 120. The homomorphism is therefore injective, and

$$|H_W| \mid \gcd(120, 28) = 4.$$

Since $|H| > 4$, the whole group cannot stabilize a line. $\square$

Now form $G = V \rtimes H$, with multiplication $(v, h)(w, k) = (v + hw, hk)$, and write $\pi : G \rightarrow H$ for projection.

LEMMA 8. *If $K \leq G$ does not contain V , then $b_f(K) \leq 2$. More precisely, if $K \cap V = 1$, then $\mu'(K) \leq 3$; if $K \cap V$ is a line, then $\mu'(K) \leq 2$.*

PROOF. Every additive subgroup of V is an $\mathbf{F}_{29}$ -subspace. If $K \cap V = 1$, projection embeds K into H , so Lemma 5 and subgroup monotonicity give both bounds. Otherwise $W = K \cap V$ is a line, normal in K , and $\pi(K) \leq H_W$ is cyclic of order dividing four. A Sylow 2-subgroup of K maps isomorphically onto $\pi(K)$ and complements W . Its action on W is faithful by Lemma 7. Lemma 6 gives $\mu'(K) \leq 2$, hence also $b_f(K) \leq 2$. $\square$

5. The invariants of the affine group

PROPOSITION 9. *One has $b_f(G) \leq b(G) \leq 3$.*

PROOF. First let $K_1, \dots, K_t$ be a meet-irredundant family with total intersection 1. Some K_i does not contain V . Lemmas 4 and 8 imply

$$t - 1 \leq b_f(K_i) \leq 2,$$

so $b_f(G) \leq 3$.

Every nontrivial normal subgroup $N \triangleleft G$ contains V . Indeed, irreducibility gives $N \cap V = 1$ or V . In the first case $[N, V] \leq N \cap V = 1$, whence $N \leq C_G(V) = V$, since the matrix action is faithful. This forces $N = 1$, a contradiction. Every nonfaithful action of G therefore factors through $G/V \cong H$, and its minimal bases have size at most $b(H) = 3$. $\square$

Let $e_1 = (1, 0)^\top$ and $e_2 = (0, 1)^\top$. The set

$$S = \{(e_1, I), (0, R_1), (0, R_2), (0, R_3)\} \quad (6)$$

is independent: projecting to H prevents any one of the three linear elements from being generated by the others, and the translation is outside the zero-translation complement. It also generates G , since the H -orbit of e_1 spans the irreducible module V .

PROPOSITION 10. *One has $\mu'(G) = 4$.*

PROOF. The set (6) gives the lower bound. For the upper bound consider a meet-irredundant family $K_1, \dots, K_t$, without any condition on its total intersection. If all members contain V , quotienting by V gives a meet-irredundant family in H , so $t \leq 3$. Otherwise choose $K_i \not\supseteq V$. Lemmas 4 and 8 give $t - 1 \leq \mu'(K_i) \leq 3$. Apply Lemma 3. $\square$

There is a faithful minimal base of size three. Define

$$\begin{aligned} L_1 &= \langle (e_1, I), (0, -I) \rangle, \\ L_2 &= \langle (e_2, I), (0, -I) \rangle, \\ L_3 &= \langle (e_1 + e_2, I), (2e_1, -I) \rangle. \end{aligned}$$

Each subgroup has order 58, and their pairwise intersections are

$$\begin{aligned} L_1 \cap L_2 &= \{1, (0, -I)\}, \\ L_1 \cap L_3 &= \{1, (2e_1, -I)\}, \\ L_2 \cap L_3 &= \{1, (-2e_2, -I)\}. \end{aligned}$$

These follow by comparing their translation lines and their cosets of elements with linear part $-I$. Their total intersection is 1. Proposition 2 gives $b_f(G) \geq 3$, realized on $G/L_1 \sqcup G/L_2 \sqcup G/L_3$, of degree 5220. Since $|G| = 29^2 \cdot 120 = 100920$, this completes the proof of Theorem 1.

6. The normality obstruction

Let $M_i = \langle S \setminus \{s_i\} \rangle$, with the elements of (6) in the displayed order. Then $M_1 = H$ and the other three subgroups are the preimages under π of the pair subgroups in (5). To see the latter assertion, each pair subgroup has order greater than four and hence cannot stabilize a line by Lemma 7; its orbit of e_1 spans V . Consequently

$$\bigcap_{i=1}^4 M_i = Z_0, \quad Z_0 = \{(0, I), (0, -I)\}.$$

This subgroup is not normal in G , since

$$(e_1, I)(0, -I)(-e_1, I) = (2e_1, -I) \notin Z_0.$$

Thus S gives a Boolean meet-semilattice of rank four with nonnormal bottom, whereas Theorem 1 excludes any such rank-four embedding with normal bottom.

Taking normal cores destroys irredundancy. The three pair subgroups have core Z in H , by simplicity of H/Z . Their preimages therefore have core $V \rtimes Z$ in G . The complement has trivial core, since every nontrivial normal subgroup of G contains V . The four cores are 1 , $V \rtimes Z$, $V \rtimes Z$, $V \rtimes Z$, whose intersection is already determined by the first member.

Reproducibility and use of artificial intelligence

The proof above is independent of subgroup enumeration. A supplementary verification package supplies exact Python and C++ implementations that reconstruct the matrix group, compare all its subgroups, and check the affine witnesses. It also records a characteristic-eleven example showing why the line-stabilizer hypothesis matters. The current package is available in the repository snapshot of 25 September 2026. The preprint and original verification materials are archived at doi:10.5281/zenodo.22929486.

OpenAI Codex (<https://openai.com/codex/>) was used during 19–25 September 2026 to develop and check the construction and proofs, generate manuscript text and exact verification programs, search and check references, and prepare the submission. Separate AI agents performed adversarial structural, computational, and source reviews. Historical application and model build identifiers were not consistently recorded. These checks are not external peer review or proof-assistant certification. The author is responsible for the mathematical claims and the submitted text.

Funding

This research received no external funding. The author declares no competing interests.

References

- [1] M. Anagnostopoulou-Merkouri and T. C. Burness, *On the regularity number of a finite group and other base-related invariants*, J. Lond. Math. Soc. **110** (2024), no. 6, e70035. <https://doi.org/10.1112/jlms.70035>.
- [2] P. J. Cameron, *Some measures of finite groups related to permutation bases*, arXiv:1408.0968 (2014). <https://arxiv.org/abs/1408.0968>.
- [3] P. J. Cameron, *Independence and bases: theme and variations*, Model Theory **3** (2024), no. 2, 417–431. <https://doi.org/10.2140/mt.2024.3.417>.
- [4] E. I. Khukhro and V. D. Mazurov (eds.), *Unsolved Problems in Group Theory: The Kourovka Notebook*, 21st edition, September 2026 update, Problem 16.45, p. 100. Official notebook and updates.

Alec Kriebel, Independent researcher, San Francisco, USA
 ORCID: 0009-0001-9320-500X
 e-mail: me@aleckriebel.com